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Research Paper

Health monitoring of plants by their emitted volatiles: A model to predict the effect of *Botrytis cinerea* on the concentration of volatiles in a large-scale greenhouse

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This paper describes a model to calculate the concentrations of (Z)-3-hexenol, α-pinene, α-terpinene, β-caryophyllene, and methyl salicylate in a greenhouse on the basis of their source and sink behaviour. The model was used to determine whether these volatile organic compounds (VOCs) can be used to indicate *Botrytis cinerea* infection in a large-scale tomato production greenhouse with a volume of 5×10^4 m^3 containing 2.2×10^4 plants. Seven experiments were done to parameterise the model for these VOCs. Based on model predictions, the *B. cinerea*-induced increase in concentration of methyl salicylate is detectable in a large-scale tomato production greenhouse when: (a) windows are fully opened, and (b) the increase continues for at least 1 h, and (c) 5% of the plants are infected. The *B. cinerea*-induced increase in concentration of methyl salicylate is also detectable when: (a) windows are closed, and (b) the increase continues for at least 6 h, and (c) 5% of the plants are infected. The *B. cinerea*-induced increase in concentration of (Z)-3-hexenol is detectable under all conditions studied. However, it is expected that besides infected plants, many additional sources of (Z)-3-hexenol exist including plant debris and nearby field crops especially upon harvest or stress. The *B. cinerea*-induced increases in concentration of α-pinene, α-terpinene and β-caryophyllene are probably undetectable in a large-scale tomato production greenhouse. Therefore, it is recommended to focus on the detection of methyl salicylate to indicate *B. cinerea* infections in large-scale tomato production greenhouses.

1. Introduction

Industries all over the world face ongoing expansion and intensification to serve the needs of a growing population. The greenhouse industry also moves towards large-scale systems while small greenhouses are closed (Breukers, Hietbrink, & Ruijs, 2008). The cultivation of crops in these large-scale greenhouses is characterised by the monoculture of high value crops at high plant density throughout the entire year. These conditions increase production per surface area.

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Nomenclature

SR_h emission rate of a VOC by healthy plants, mol s^{-1}
SR_B emission rate of a VOC by *Botrytis cinerea* infected plants, mol s^{-1}
SR_{air_in} introduction rate of a VOC together with air entering the greenhouse, mol s^{-1}
SR_{mat} emission rate of a VOC by greenhouse material, mol s^{-1}
SR_{air_out} removal rate of a VOC by air leaving the greenhouse, mol s^{-1}
SN_{mat} adsorption/absorption rate of a VOC by greenhouse materials, mol s^{-1}
SN_{water} transfer rate of a VOC into water bodies, mol s^{-1}
SN_{plant} adsorption/absorption rate of a VOC by plants, mol s^{-1}
SN_{O_3} gas-phase reaction rate of a VOC with O_3, mol s^{-1}
V volume of greenhouse, m^3
$[VOC]_{gr}$ molar concentration of a VOC in the greenhouse air, mol m^{-3}
$[O_3]_{gr}$ molar concentration of O_3 in the greenhouse air, mol m^{-3}
t time, s
n_h number of healthy tomato plants
Φ_h emission flux density of a VOC by healthy plants, mol m^{-2} s^{-1}
n_B number of *Botrytis cinerea* infected plants
Φ_B emission flux density of a VOC by *Botrytis cinerea* infected plants, mol m^{-2} s^{-1}
A_{plant} one-sided leaf area per tomato plant, m^2
f_{air_in} volumetric flow rate of air entering the greenhouse, m^3 s^{-1}
$[VOC]_{air\_in}$ molar concentration of a VOC in the incoming air, mol m^{-3}
f_{air_out} volumetric flow rate of air leaving the greenhouse, m^3 s^{-1}
$[VOC]_{air\_out}$ molar concentration of a VOC in the air leaving the greenhouse, mol m^{-3}
k_{mat_air} mean emission rate of a VOC by greenhouse materials, mol m^{-2} s^{-1}
k_{air_mat} mean adsorption/absorption rate of a VOC by greenhouse materials, mol m^{-2} s^{-1}
A_{mat} adsorption/absorption area of the greenhouse materials, m^2
A_{water} transfer area of the water bodies, m^2
$[VOC]_{water}$ molar concentration of a VOC in the water, mol m^{-3}
k_{air_water} transfer coefficient for uptake of a VOC into water bodies, m s^{-1}
k_{plant} mean adsorption/absorption rate of a VOC by plants, mol s^{-1} m^{-2}
k_{O_3} rate constant for the reaction of O_3 with a VOC, m^3 mol^{-1} s^{-1}
$[CO_2]_{gr}$ molar concentration of CO_2 in the greenhouse, mol m^{-3}
$[CO_2]_{amb}$ molar concentration of CO_2 in ambient air outside the greenhouse, mol m^{-3}

However, the year round production of one single, high density crop also provides excellent circumstances to establish and spread pathogens and pests throughout the greenhouse (Elad, 1999; Lenteren, 2000).

Nowadays, regular human inspection is common practice to monitor crops for the presence of diseases and pests. These inspections must be accurate in large-scale greenhouses since inaccurate inspections allow local disease or pest problems to disperse rapidly over long distances which in turn result in large economic losses. These accurate, on-site inspections of crops are time-consuming and require skilled personnel which in turn leads to high costs. As a result, greenhouse managers want to automate these on-site inspections to limit the demand for manual labour. Thus, expansion and intensification of the greenhouse industry increase the demand for an automated system to monitor greenhouse crops for the presence of diseases and pests. Such a system would facilitate immediate actions and prevent further spread by controlling the problem right at the source.

Researchers have sought efficient ways to monitor greenhouse crops for the presence of diseases and pests. One option is the analysis of air to identify and/or quantify trace levels of volatile organic compounds (VOCs) emitted from plants which suffer from pest or disease problems (Schütz & Weißbecker, 2003).

Infections of crops with the pathogenic fungus *Botrytis cinerea* are among the most common cause of reduced yields in greenhouse horticulture (Elad & Stewart, 2004). During 2005–2008, we carried out a broad range of experiments to test whether VOCs can be used as an indicator of the presence of *B. cinerea* infection in a large-scale tomato production greenhouse. In the first phase, we analysed the air surrounding tomato leaves individually enclosed in 300 ml Petri dishes. Results demonstrated that leaf-emitted VOCs can be used as an indicator of the presence of *B. cinerea* infection at this ultra-small scale (Thelen et al., 2006). In the second phase, we analysed the air surrounding several tomato plants grown in a 1 m^3 chamber. Results of that follow-up study demonstrated that plant-emitted VOCs can be used as an indicator of the presence of *B. cinerea* infection also at this intermediate scale (Jansen, Miebach, Kleist, van Henten, & Wildt, 2009). In the third phase, we analysed the air surrounding 60 tomato plants, grown in a small, 270 m^3 greenhouse. Results obtained during that phase indicate that crop-emitted VOCs can be used as an indicator of the presence of *B. cinerea* infection in a small-scale greenhouse (Jansen et al., in press). The 60 plants grown in the small-scale greenhouse were situated on a 42 m^2 floor area. In commercial large-scale greenhouses, many more plants are grown on far larger floor areas. For example, at present, the majority of such large-scale greenhouses in Western European countries, such as the Netherlands, may contain 2×10^4–2×10^5 plants and have floor areas which range between 10^4 and 10^5 m^2 (van Henten, 2006). The objective of this study was to determine whether plant-emitted volatiles can be used to indicate *B. cinerea* infection in a large-scale greenhouse. In this research this objective was addressed with a modelling approach. Though, in the future, measurements in a real greenhouse should confirm the findings, it was considered a valuable first step to scale-up the results obtained in a small-scale greenhouse to the size of current commercial large-scale greenhouses in order to assess the potential of VOC-based crop health monitoring.

2. Materials and methods

2.1. Experimental greenhouse

A schematic illustration of the small-scale greenhouse used in this study is presented in Fig. 1. This greenhouse has been described in detail by Körner, van't Ooster, and Hulsbos (2007). In short, the floor area of the greenhouse is 44 m^2 and the total volume including the basement underneath is 270 m^3. A fan located in the basement was used to maintain a constant internal air circulation of 2×10^4 m^3 h^{-1}. The temperature in the greenhouse was set at 22/16 ± 1.0 °C day/night. The relative humidity inside the greenhouse was set at about 70/90 ± 5% day/night. An air conditioner, consisting of a heater, cooler, and dehumidifier situated in the basement, maintained these temperatures and humidity. The greenhouse was semi-closed; it did not contain any windows that could be opened. However, air might have entered through leaks in the basement, and might have left the greenhouse through leaks

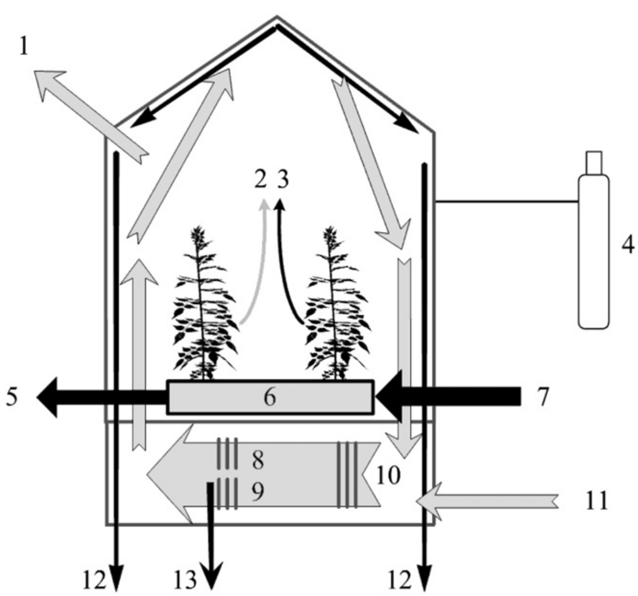

Fig. 1 – Schematic illustration of the small-scale greenhouse used in this study: (1) air leaving the greenhouse, (2) emission of volatile organic compounds (VOCs) by the plants, (3) transpiration by the plants, (4) CO_2-supply, (5) water leaving the slab, (6) slab, (7) water entering the slab, (8) cooler, (9) dehumidifier, (10) heater, (11) air entering the greenhouse, (12) water leaving the gutters, and (13) water leaving the dehumidifier. Grey arrows represent the transport of airborne VOCs, and black arrows represent the transport of liquid water.

in the glass cover. Pure CO_2 was supplied to maintain a CO_2 concentration of 420 μmol mol^{-1}.

2.2. Mass transfer model

For this study it was assumed that the air in the greenhouse is perfectly mixed. This means that the concentration of plant-emitted VOCs was assumed to be similar at each location within the greenhouse. It was furthermore assumed that the volumetric flow rate of air entering the greenhouse due to forced or natural ventilation was equal to the volumetric flow rate of air leaving the greenhouse. A certain number of healthy plants is assumed emitting VOCs constitutively. Another number of plants was assumed to have *B.cinerea*-induced emissions. The leaf area per plant was assumed to be the same.

Four sources and five sinks were considered as most relevant for the mass balance of VOCs in greenhouse air. The first source was the emission of a VOC by healthy plants. The emission rate of a VOC by healthy plants (SR_h) gives the total amount of the VOC introduced by the plants into the greenhouse air per unit of time. It depends on the number of healthy plants (n_h), on the emission flux density of the VOC by the healthy plants (Φ_h), and on the one-sided leaf area per plant (A_{plant}). This rate is described by Eq. (1):

$$SR_h = n_h \cdot \Phi_h \cdot A_{plant} \quad (1)$$

The second source was the emission of a VOC by *B. cinerea* infected plants. The emission rate of a VOC by *B. cinerea* infected plants (SR_B) depends on the number of infected plants (n_B), on the emission flux density of the VOC by *B. cinerea* infected plants (Φ_B), and on the one-sided leaf area per plant (A_{plant}). This rate was described by Eq. (2):

$$SR_B = n_B \cdot \Phi_B \cdot A_{plant} \quad (2)$$

The third source was the introduction of a VOC together with the air entering the greenhouse. For instance, trees near a greenhouse may emit some of the VOCs considered in this study, which can then be transferred by wind to the ambient air outside the greenhouse, and can then enter the greenhouse. The introduction rate of a VOC by this process (SR_{air_in}) was assumed to depend on the volumetric flow rate of this air (f_{air_in}), and the concentration of the VOC in this air ($[VOC]_{air\_in}$). This rate was described by Eq. (3):

$$SR_{air\_in} = f_{air\_in} \cdot [VOC]_{air\_in} \quad (3)$$

The fourth source was the emission of a VOC by greenhouse materials; for example, glues or epoxies, drying agents in paints, and soft plastics emit significant amounts of VOCs into the surrounding air (e.g. Guo, 2002). The emission rate of a VOC by greenhouse materials (SR_{mat}) was assumed to depend on the emission coefficient of greenhouse materials (k_{mat_air}), and the transfer area of the greenhouse materials (A_{mat}). This rate was given by Eq. (4):

$$SR_{mat} = k_{mat\_air} \cdot A_{mat} \quad (4)$$

The first sink was the removal of a VOC by air leaving the greenhouse. For instance, open windows at the top of a greenhouse allow air to leave together with the VOCs therein. The removal rate of a VOC by this process (SN_{air_out}) depends on the volumetric flow rate of air leaving the greenhouse (f_{air_out}), and the concentration of the VOC in this air ($[VOC]_{air\_out}$). This rate was described by Eq. (5):

$$SN_{air\_out} = f_{air\_out} \cdot [VOC]_{air\_out} \tag{5}$$

The second sink was the adsorption/absorption of a VOC by greenhouse materials. This process might be important since many different building materials absorb VOCs (see Yang et al., 2001 and references therein). The adsorption/absorption rate of a VOC by greenhouse materials (SN_{mat}) was assumed to depend on the adsorption/absorption coefficient of greenhouse materials (k_{mat_air}), and the transfer area of the greenhouse materials (A_{mat}). This rate was described by Eq. (6):

$$SN_{mat} = k_{air\_mat} \cdot A_{mat} \tag{6}$$

The third sink was the adsorption and/or absorption of a VOC by plants. This sink can be the result of adsorption of VOCs on the plant cuticle as well as absorption through the stomata (Riederer, Daiβ, Gilbert, & Köhle, 2002; Seco, Peñuelas, & Filella, 2007). The adsorption/absorption rate of a VOC by plants (SN_{plant}) was assumed to depend on the number of healthy plants (n_h), the number of *B. cinerea* infected plants (n_B), the adsorption/absorption coefficient of plants (k_{plant}), and the one-sided leaf area per plant (A_{plant}). This rate was described by Eq. (7):

$$SN_{plant} = (n_h + n_B) \cdot k_{plant} \cdot A_{plant} \tag{7}$$

The fourth sink was the degradation of VOCs through gas-phase reactions. Such gas-phase reactions between plant-emitted VOCs and hydroxyl radicals (OH), nitrate radicals (NO_3) and ozone (O_3) are common in the troposphere (Atkinson & Arey, 2003; Canosa-Mas, Duffy, King, & Thompson, 2002). Several VOCs, including α-terpinene and β-caryophyllene are therefore difficult to detect in the troposphere since they have a lifetime of respectively 0.5 and 2 min as a result of gas-phase reactions (Atkinson & Arey, 2003).

Degradation of VOCs through gas-phase reactions with O_3 was regarded as most important. The removal rate of a VOC by gas-phase reaction with O_3 (SR_{O3}) was assumed to depend on the volume of the greenhouse (V), the concentration of the VOC in the greenhouse air ($[VOC]_{gr}$), the concentration of O_3 in the greenhouse air ($[O_3]_{gr}$) and the rate constant for the reaction of O_3 with the VOC (k_{O_3}). This dependence was described by Eq. (8).

$$SN_{O_3} = V \cdot [VOC]_{gr} \cdot [O_3]_{gr} \cdot k_{O_3} \tag{8}$$

The fifth sink was the transfer of VOCs into water bodies. This may occur if a VOC dissolves in liquid water through contact with airborne VOCs, such as for condensation droplets on the inner side of the greenhouse cover. This sink seems relevant for VOCs with a high Henry's law coefficient representing high solubility of the VOC in water. The transfer rate of a VOC into water bodies (SN_{water}) was assumed to depend on the transfer coefficient for the uptake of a VOC into a water body (k_{air_water}), the transfer area of the water body (A_{water}), and the difference in concentration between the VOC in greenhouse air ($[VOC]_{gr}$) and the VOC in the water body ($[VOC]_{water}$). This rate was described by Eq. (9).

$$SN_{water} = k_{air} \cdot A_{water} \cdot \left([VOC]_{gr} - [VOC]_{water}\right) \tag{9}$$

Assuming the above given sources and sinks to be the dominant ones, the time course of the concentration of a VOC was described by Eq. (10) in which the individual sources and sinks constitute the mass balance of the system.

$$V \cdot \frac{d[VOC]_{gr}}{dt} = SR_h + SR_B + SR_{air\_in} + SR_{mat} - SN_{air\_out} - SN_{mat} - SN_{water} - SN_{plant} - SN_{O_3} \tag{10}$$

3. Experimental design

3.1. Experiment 1: introduction of VOCs together with air entering a greenhouse

Air samples were collected outside a small-scale greenhouse (Fig. 1) to estimate the entrance rate of VOCs by air entering this greenhouse. The method to derive the concentrations of VOCs in the ambient air outside the greenhouse was based on active sampling and GC–MS analysis. This method has been described in detail by Jansen, Hofstee, et al. (2009) and provides detection limits in the order of 1–10 pptv (parts per trillion by volume, 10^{-12}). The concentrations of VOCs in the ambient air outside the greenhouse were assumed to be equal to the concentrations of VOCs in the air entering the greenhouse ($[VOC]_{air\_in}$). These concentrations were substituted into Eq. (3) to calculate the entrance rates of VOCs with air entering this greenhouse.

3.2. Experiment 2: removal of VOCs by air leaving a greenhouse

The removal of VOCs by air leaving a greenhouse depends on the volumetric flow rate of this air. The flow rate was determined for the small-scale greenhouse (Fig. 1 but without plants), using the tracer gas concentration decay technique (Bot, 1983). This technique is based on the decrease in concentration of a tracer gas inside and the nearly constant concentration of this tracer gas in the ambient air outside the greenhouse. In this study, CO_2 was used as a tracer gas. The volumetric flow rate of air leaving the greenhouse (f_{air_out}) was calculated by Eq. (11). This flow rate was substituted into Eq. (5) to calculate the removal rates of VOCs by air leaving the greenhouse. The tracer gas experiment was replicated three times.

$$[CO_2]_{gr}(t) = [CO_2]_{amb} + \left([CO_2]_{gr,t=0} - [CO_2]_{amb}(t)\right) \cdot \exp\left(-\frac{f_{air\_out}}{V} t\right) \tag{11}$$

3.3. Experiment 3: emissions of VOCs by greenhouse materials

The air inside the small-scale greenhouse (Fig. 1 but without plants) was sampled to estimate whether the materials of this greenhouse emit any of the investigated VOCs. The method to collect these samples and to analyse them followed the procedure described by Jansen, Hofstee, et al. (2009). This experiment was replicated three times.

3.4. Experiment 4: adsorption/absorption of VOCs by greenhouse materials

To estimate whether greenhouse materials adsorb and/or absorb VOCs, we evaporated (Z)-3-hexenol, methyl salicylate, α-pinene, α-terpinene and β-caryophyllene (GC grade; Fluka, Milwaukee, WI, USA) inside the small-scale greenhouse (Fig. 1 but without plants). The decays in concentration of these VOCs were then compared to the decay of the CO_2 tracer gas. This method could only be used because the internal air circulation inside the greenhouse was large (2×10^4 m^{-3} h^{-1}) which resulted in well mixed air. In order to evaporate the VOCs, we transferred about 10 ml of each VOC into individual 25 ml glass vials which were immediately capped. The capped vials were then put in the centre of the greenhouse for at least 50 min to stabilise their temperature. After this period, the vials were uncapped to start evaporation. After 60 min, the vials were capped to end evaporation. Air samples were collected before vials were uncapped and 1, 2, 4, 7, 11, 16 and 24 h after the vials were capped and analysed as described by Jansen, Hofstee, et al. (2009). This experiment was replicated six times with a time interval of at least 48 h.

3.5. Experiment 5: adsorption/absorption of (Z)-3-hexenol by tomato plants

A laboratory set-up as described in detail by Heiden, Kobel, Langebartels, Schuh-Thomas, and Wildt (2003) was used to determine the adsorption/absorption rate of (Z)-3-hexenol by tomato plants (*Lycopersicon esculentum* Mill). Tomato plants were only tested for adsorption and/or absorption of (Z)-3-hexenol since the other VOCs considered in this study are emitted from tomato plants enclosed in the set-up (see Jansen, Miebach, et al. (2009) and references therein), which makes it impossible to distinguish between adsorption, absorption and emission.

The tomato plants used for this experiment were six or eight weeks old with leaf areas per plant between 0.03 and 0.1 m^2. Briefly, the shoot part of such a plant was enclosed in a 1.1 m^3 glass chamber. The light intensity at the top of the plant was set at a photosynthetic photon flux density (PPFD) of 480 μmol photons m^{-2} s^{-1}. The temperature inside the chamber was set at 20 °C, the humidity inside the chamber was set at 70% relative humidity and the CO_2 concentration inside the chamber was maintained at 350 μmol mol^{-1} to mimic natural conditions. Clean and moistened air, supplemented with CO_2, was entering this chamber at a flow rate of 30 l min^{-1}. After enclosure of the shoot part, about 1.1×10^{-10} mol s^{-1} of (Z)-3-hexenol was added into this incoming air. Concentrations of (Z)-3-hexenol were alternatively measured in the air entering and in the air leaving the chamber. This experiment was replicated twice with different tomato plants.

3.6. Experiment 6: gas-phase reactions of VOCs with O_3

Colorimetric detector tubes (Kitagawa, Japan) were used to measure the concentration of O_3 in the air inside the small-scale greenhouse (Fig. 1). At the time of measurement, the greenhouse contained 60 tomato plants (*L. esculentum* Mill) of about 1.8 m in height. The concentration of O_3 was measured in two independent replicates. For each replicate, 4.8 l of air was sucked through the tube.

3.7. Experiment 7: the transfer of VOCs into water bodies

Neither the transfer areas of water bodies nor the transfer coefficient are predictable by simple theories or determinable by straightforward measurements since both depend on a complex array of hydrodynamic and aerodynamic factors. We therefore conducted an experiment with plants in the greenhouse to simulate realistic conditions. These plants served as a source of water as a result of condensation of transpired water onto cold surfaces. Hence, in this experiment, an overall loss was measured:

$$\sum \text{Loss} = SN_{air\_out} + SN_{mat} + SN_{water} + SN_{plant} + SN_{O_3} \tag{12}$$

The results of experiments 4, 5 and 6 indicated that greenhouse materials, tomato plants, and gas-phase reactions are negligible sinks for VOCs and therefore, Eq. (12) reduces to Eq. (13):

$$\sum \text{Loss} = SN_{air\_out} + SN_{water} \tag{13}$$

To estimate the losses described in Eq. (13), we evaporated the VOCs considered in this study inside the small-scale greenhouse, about 1.8 m long containing 60 tomato plants (*L. esculentum* Mill). The evaporation of VOCs, the sampling of air, and the analyses of samples followed the procedure as described for experiment 4. Assuming that the loss of the VOC by solution in the water bodies is proportional to the concentration of the VOC in the greenhouse air, this loss process is of first order. As the loss by removal with the air leaving the greenhouse is also a first order process, the decay can be described by Eq. (14).

$$[VOC]_{gr}(t) = \left([VOC]_{gr,t=0}\right)\cdot\exp\left[\frac{f_{airout} + (k_{air\_water}\cdot A_{water})}{V}t\right] \tag{14}$$

The decrease in concentration after evaporation was fitted to an exponential decay curve to calculate the term $k_{air\_water}\cdot_{water}$.

Two different tomato crops were used to determine $k_{air\_water}\cdot_{water}$ per VOC. Crop *A* was planted in February 2008 and crop *B* was planted in August 2008. For crop *A*, VOCs were evaporated first at 72, a second time at 74, and a third time at 76 days after planting. For crop *B*, the VOCs were evaporated at 76, and a second time at 78 days after planting.

4. Results and discussion

4.1. Experiment 1: introduction of VOCs together with air entering a greenhouse

None of the VOCs considered in this study could be detected in the ambient air outside the greenhouse given the detection limit of our system of 1–10 pptv for these VOCs (Jansen, Hofstee, et al., 2009). The entrance of VOCs by air entering a greenhouse was therefore considered negligible as a source and the term SR_{air_in} was neglected in the further analysis below.

The absence, or at least very low concentrations, of these VOCs in ambient air was counter-intuitive since nearby vegetation such as trees, hedges, and field crops are potential sources of VOCs (Army et al., 1991), especially near harvest (e.g. Davison et al., 2008). We assume that the amount of vegetation present near the greenhouse was insufficient to induce considerable emissions of VOCs. Also crops inside greenhouses located near our experimental facility would be possible sources of VOCs but the number and size of nearby greenhouses were probably too small to play an important role. However, these sources remain relevant for large-scale greenhouse production because they are, in general, surrounded by many other large-scale greenhouses. The absence of relevant VOCs may also be explained by gas-phase reactions in the troposphere in which VOCs are degraded. Such degradation seems plausible since the VOCs used in this study have tropospheric lifetimes in the order of minutes to hours (Atkinson & Arey, 2003; Canosa-Mas et al., 2002). The tropospheric lifetime, together with wind direction and wind speed are important characteristics to consider since they determine the distance VOCs can move before degradation. In future studies, the air surrounding such large-scale greenhouses should be analysed to give an indication of VOC concentrations in this air. This should preferably be done at different times of the year and different times of the day due to the seasonal and diurnal rhythm of VOC emissions by vegetation (e.g. Tarvainen et al., 2005).

4.2. *Experiment 2: removal of VOCs by air leaving a greenhouse*

The tracer gas experiment resulted in an exponential decrease of the CO_2 tracer concentration inside the small-scale greenhouse. Based on Eq. (11) the flow rate of air leaving the greenhouse (f_{air_out}) was 0.041 ± 0.001 m^3 s^{-1}. This flow rate was substituted into Eq. (5) to calculate the removal of VOCs by air leaving the greenhouse.

The removal rate of VOCs by air leaving the greenhouse turned out to be the most dominant sink for α-pinene, α-terpinene and β-caryophyllene. Whether or not windows are opened will have a large effect on this removal rate. The windows of current large-scale commercial greenhouse are often closed, but also regularly opened to dehumidify the greenhouse, to cool the greenhouse, and to supply extra CO_2. Fluctuations in VOC concentrations caused by closing and opening of windows may then be misinterpreted. The effect of this procedure should be studied in more detail to find out whether this obstructs the interpretation of plant VOC concentrations. Trials are currently ongoing to test the effect of different window configurations on VOC concentrations in greenhouse air.

4.3. *Experiment 3: emissions of VOCs by greenhouse materials*

None of the VOCs considered in this study could be detected in any of the air samples collected inside the small-scale greenhouse without plants. The emission of VOCs by greenhouse materials was therefore neglected.

This seems justifiable since the main material used in a modern greenhouse is glass which emits almost no VOCs and therefore this source process is more or less irrelevant. Other potential sources include plastics but the VOCs emitted from plastics are different from those from plants, so this should also not be a problem.

4.4. *Experiment 4: adsorption/absorption of VOCs by greenhouse materials*

In the greenhouse without plants, the decays in concentrations of (Z)-3-hexenol, α-pinene and methyl salicylate were similar to the decay in the concentration of the CO_2 tracer gas. As an example, the decay in concentration of (Z)-3-hexenol and the decay in concentration of the CO_2 tracer gas are shown in Fig. 2. The similarity of the decay curves implies that in the greenhouse without plants, these VOCs were mainly removed by air leaving the greenhouse. Referring to Fig. 2, it can be seen that the concentration of (Z)-3-hexenol does not decrease to zero. Although (Z)-3-hexenol was undetected in the ambient air (experiment 1), small amounts of (Z)-3-hexenol may have entered the greenhouse during execution of experiment 4.

Therefore, it was assumed that adsorption/absorption of these VOCs by greenhouse materials could be neglected in the further analysis. In the greenhouse without plants, the decays in concentration of α-terpinene and β-caryophyllene were faster; maybe as a result of oxidation on metal surfaces. This result demonstrates that these processes should not be neglected completely since many researchers have demonstrated that several materials have the capacity to absorb and/or adsorb VOCs (Jorgenson, 1999; Singer, Revzan, Hotchi, Hodgson, & Brown, 2004). However, glass is one of the main materials used in a modern greenhouse. Most chemicals hardly absorb and/or adsorb onto glass materials which justifies the assumption that this sink process is less relevant.

4.5. *Experiment 5: adsorption/absorption of (Z)-3-hexenol by tomato plants*

In none of the experiments in which tomato plants were exposed to (Z)-3-hexenol were significant differences found between the concentration of (Z)-3-hexenol in the air entering

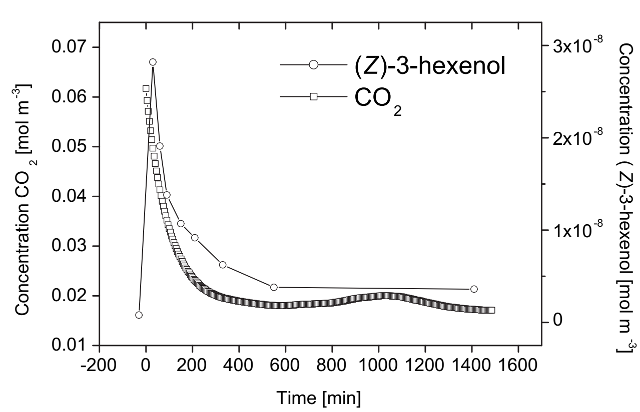

Fig. 2 – Decay in concentration of (Z)-3-hexenol and CO_2 tracer gas. Decay curves were obtained in the small-scale greenhouse without plants.

and in the air leaving the chamber (data not shown). The adsorption/absorption of (Z)-3-hexenol by tomato plants was therefore neglected. It needs to be stressed that, based on the experiments we did, we cannot exclude that tomato plants adsorb and/or absorb methyl salicylate, α-pinene, α-terpinene or β-caryophyllene. But, in steady-state conditions, adsorption/absorption of VOCs will not be an important loss process as the net flux of desorbed and adsorbed VOCs will be zero, unless there is a large sink inside the plant due to chemical degradation (Copolovici, Filella, Llusià, Niinemets, & Peñuelas, 2005).

4.6. Experiment 6: gas-phase reactions of VOCs with O_3

In both experiments, the concentration of O_3 inside the small-scale greenhouse containing 60 tomato plants was below the detection limit of the method (<2 ppb). Therefore, gas-phase reactions of VOCs with O_3 (SN_{O_3}) were neglected.

This result was expected since plants take up O_3 very efficiently (e.g. Loreto & Fares, 2007; Neubert, Kley, Wildt, Segschneider, & Förstel, 1993) and thus, concentrations of O_3 in a greenhouse with high plant density should be very low. Therefore, gas-phase reactions of plant-emitted VOCs with O_3 inside the greenhouse will be negligible as a sink, which explains the presence of these VOCs in greenhouse air (Jansen et al., in press; Jansen, Hofstee, et al., 2009; Jansen, Miebach, et al., 2009; Jansen, Takayama, et al., 2009).

4.7. Experiment 7: the transfer of VOCs into water

After vials were capped, the concentration of VOCs gradually decreased. Not unexpectedly, the decays were most rapid for (Z)-3-hexenol and methyl salicylate since they have high Henry's law coefficients of 0.33 and 0.75 mol m^{-3} Pa^{-1} respectively (Atlan, Trelea, Saint-Eve, Souchon, & Latrille, 2006; Karl, Guenther, Turnipseed, Patton, & Jardine, 2008). Decays were slow for α-pinene, α-terpinene and β-caryophyllene, probably because they have low Henry's law coefficients of 7.38×10^{-5}, 2.87×10^{-4} and 3.98×10^{-3} mol m^{-3} Pa^{-1} respectively (Copolovici & Niinemets, 2005; Weschler & Nazaroff, 2008). Fig. 3 provides an example of the decays in concentrations of (Z)-3-hexenol, β-caryophyllene and methyl salicylate after vials were capped to stop evaporation. Based on the decays in concentration, the transfer into water bodies ($k_{air\_water} \cdot A_{water}$) was calculated (Table 1).

Table 1 – Calculated $k_{air\_water} \cdot A_{water}$ values for crop A and crop B.

Compound	$k_{air\_water} \cdot A_{water}$ [m^3 s^{-1}]					Average (sd)
	Crop A			Crop B		
(Z)-3-Hexenol	0.41	0.41	0.15	0.13	0.14	0.25 (0.15)
α-Pinene	0.00	0.01	0.02	0.00	0.03	0.01 (0.01)
α-Terpinene	0.07	0.06	0.05	0.04	0.06	0.06 (0.01)
β-Caryophyllene	0.02	0.02	0.02	0.00	0.02	0.01 (0.01)
Methyl salicylate	0.04	0.05	0.04	0.06	0.06	0.05 (0.01)

These results suggest that transfer of VOCs into water is an important sink strength for the water soluble VOCs (Z)-3-hexenol and methyl salicylate. These two VOCs are possible indicators for *B. cinerea* infection of tomato (see Table 2). It is therefore of utmost importance to study this sink in more detail to determine whether or not these VOCs are detectable under conditions that occur in large-scale greenhouses.

4.8. Model predictions for large-scale greenhouses

Based on the results of experiments 1–7, Eq. (10) was reduced to:

$$V \cdot \frac{d[VOC]_{gr}}{dt} = SR_h + SR_B - SN_{air\_out} - SN_{water} \quad (15)$$

Eq. (15) was implemented in Matlab™ (Release 14; The MathWorks Inc., MA, USA) to predict the effects of parameter changes on the concentrations of VOCs in a large-scale greenhouse under four different scenarios. The main quantities determining these concentrations are the volume of the greenhouse, the emission flux densities of the individual VOCs, the total number of plants, the fraction of *B. cinerea* infected plants, the leaf area per plant, and the exchange of air with the outdoor environment.

Emission flux densities of VOCs by healthy and infected plants are required to calculate their emission rates. Table 2 provides the emission flux densities used in this study. These data were taken from our previous work in which tomato plants were severely infected with *B. cinerea* (Jansen, Miebach, et al., 2009).

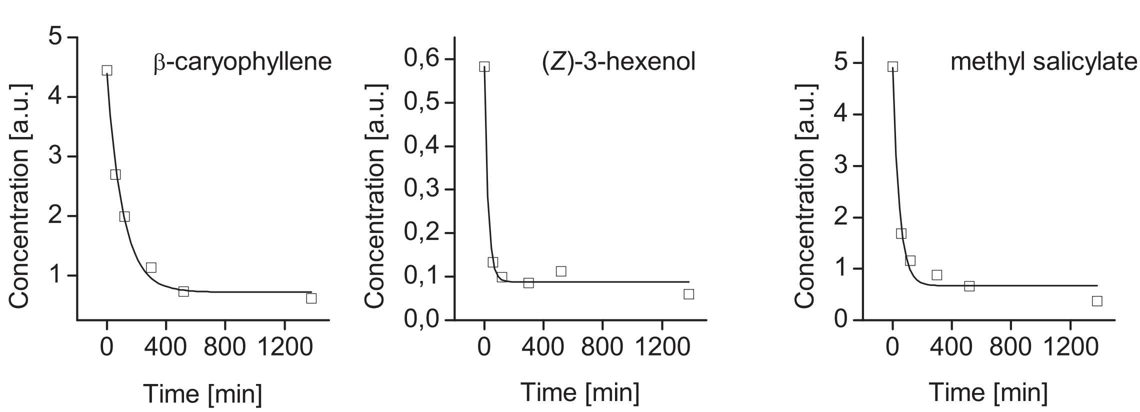

Fig. 3 – Decays in concentrations of β-caryophyllene, (Z)-3-hexenol, and methyl salicylate in greenhouse air after vials were capped to stop evaporation. Decay curves were obtained in the small-scale greenhouse with plants.

Table 2 – Emission flux densities of the volatile organic compounds used in this study. Data were taken from our previous work in which tomato plants were severely infected with *Botrytis cinerea* (Jansen, Miebach, et al., 2009; Jansen, Takayama, et al., 2009).

Compound	Emission flux density [nmol m^{-2} s^{-1}]	
	Healthy	*B. cinerea* infected
(Z)-3-Hexenol	0	3.1×10^{1}
α-Pinene	2.1×10^{-2}	3.0×10^{-2}
α-Terpinene	2.5×10^{-2}	3.9×10^{-2}
β-Caryophyllene	1.1×10^{-3}	1.9×10^{-3}
Methyl salicylate	1.1×10^{-1}	3.2×10^{-1}

Four scenarios were studied. For each scenario, values were set for the volume of the greenhouse, the total number of tomato plants, the proportion of *B. cinerea* infected tomato plants, the period of time in which the *B. cinerea*-induced increase in emission of VOCs from a certain proportion of plants is above the baseline level emission of healthy plants, the leaf area per plant, and the flow rate of air leaving the greenhouse (Table 3).

Eq. (15) was used to predict the baseline and *B. cinerea*-induced concentrations of VOCs in the greenhouse atmosphere under the four different scenarios listed in Table 3. For these scenarios, starting at an initial concentration of 0 mol m^{-3}, the emissions from the healthy plants (SR_h) led to an increase of VOC concentrations until a steady-state concentration was reached. These steady-state concentrations were determined by the sinks and sources given in Eq. (15) with $SR_B = 0$. After steady state was reached, it was assumed that a given proportion of plants (Table 3) became infected by *B. cinerea*: in this case all at the same time, $t = 1500$ min. As a consequence, the emission of infected plants increased. This led to an increase in VOC concentrations in the greenhouse air which in magnitude depended on the scenario used. For our model we assumed that an atypical increase in the concentration of stress-induced VOCs will alert the grower to the presence of stress. Then, proper measures might be undertaken by the grower to reduce this stress. Such measures will result in a reduced emission of stress-induced VOCs to low background levels. As an example, in Fig. 4 we show the predicted concentrations of methyl salicylate under scenarios 1, 2, 3, and 4. Fig. 4 shows that it may take several hours to reach steady-state conditions after introducing the plants. However, steady-state conditions are no precondition to detect stress-induced plant responses. Steady-state conditions are just helpful because it is easier to detect the increase of stress-induced emissions under such conditions.

Table 3 – Volume of the greenhouse (*V*), total number of tomato plants, proportion of *Botrytis cinerea* infected tomato plants, the excitation period, the leaf area per plant (A_{plant}), and the flow rate of air leaving the greenhouse (f_{air_out}) for scenarios 1, 2, 3, and 4.

	Scenarios			
	1	2	3	4
Volume [m^3][a]	5×10^4	5×10^4	5×10^4	5×10^4
Total number of plants[b]	2.2×10^4	2.2×10^4	2.2×10^4	2.2×10^4
B. cinerea infected plants [%]	0.5	5	0.5	5
Excitation period [min]	60	60	60	60
Leaf area per plant [m^2]	1	1	1	1
Flow rate of air leaving the Greenhouse [m^3 s^{-1}]	4.6[c]	4.6[c]	278[d]	278[d]

a Volume is based on a floor area of 10^4 m^2 and a height of the greenhouse of 5 m; these dimensions are common for Dutch greenhouses.
b Total number of tomato plants corresponding to the given greenhouse volume and floor area.
c This flow rate is equal to an exchange of one-third of the greenhouse volume per hour; a reasonable value for modern Dutch greenhouses when windows are closed.
d This flow rate is equal to an exchange of 20-times the greenhouse volume per hour; a reasonable value for modern Dutch greenhouses when windows are fully opened.

Detection of *B. cinerea* infections through plant-emitted VOCs requires analytical instruments with high accuracy, low detection limits and good resolution. The VOCs considered in this study were measured routinely with an accuracy expressed as relative standard deviation (RSD) of 10%, detection limits of 1 nmol m^{-3}, and a time resolution of 1 h. For instance, Greenberg, Lee, Helmig, and Zimmerman (1994) used an instrument that consisted of cryo-trapping and gas chromatography–flame ionisation detection to detect VOCs with an accuracy of *ca*. 10% RSD at a detection limit of less than 1 nmol m^{-3}. An accuracy of 10% RSD and a detection limit of 1 nmol m^{-3} was therefore used to estimate whether the *B. cinerea*-induced increase in concentration of (*Z*)-3-hexenol, α-pinene, α-terpinene, β-caryophyllene, and methyl salicylate (Table 4) is detectable.

Based on the above-mentioned accuracy and detection limit, the *B. cinerea*-induced increase in concentration of (*Z*)-3-hexenol is detectable under all scenarios. In contrast, the *B. cinerea*-induced increase in the concentration of α-pinene, α-terpinene and β-caryophyllene is insufficient to be detectable. The *B. cinerea*-induced increase in concentration of methyl salicylate is only detectable under scenario 4, i.e. 5% of the plants infected and windows fully opened. It should be mentioned that the relative increase in concentration of methyl salicylate under scenario 4 is not easy to detect because this increase is just above the maximum attainable accuracy of 10% RSD. The estimates of the detectability of *B. cinerea*-induced increase in concentration of (*Z*)-3-hexenol, α-pinene, α-terpinene, β-caryophyllene, and methyl salicylate are summarised in Table 5.

Future sensitivity analysis should include the period of time in which the *B. cinerea*-induced increase in emission of VOCs from a certain proportion of plants is above the baseline level emission of healthy plants. For instance, the *B. cinerea*-induced increase in concentration of methyl salicylate is also detectable under scenario 2, i.e. 5% of the plants infected and windows closed, if this period of time increases from 60 to at least 360 min (data not shown). Furthermore, a sensitivity analysis should include the emission flux densities of methyl salicylate by healthy tomato plants. These values were obtained under laboratory conditions and should be confirmed under undisturbed greenhouse conditions.

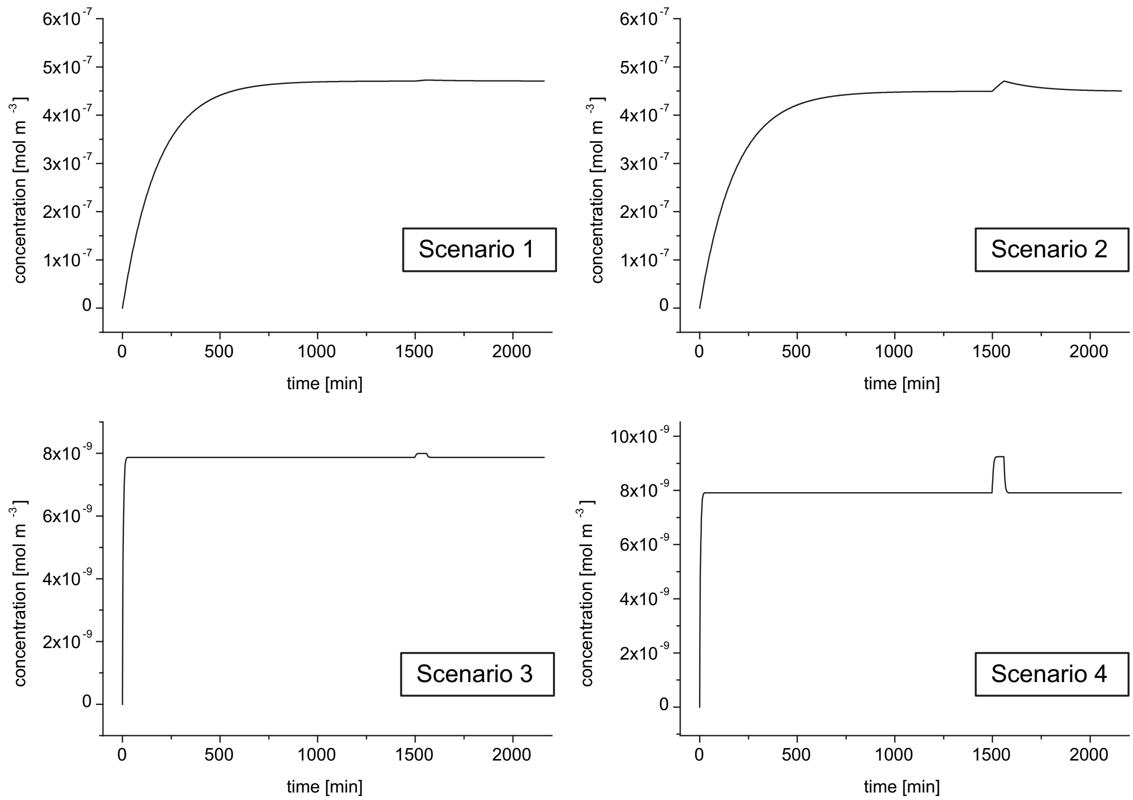

Fig. 4 – The predicted baseline and *Botrytis cinerea*-induced concentrations of methyl salicylate under scenarios 1, 2, 3, and 4. The scenarios are presented in Table 3. At t = 0 min, healthy plants are introduced into the greenhouse. At t = 1500 min, a pulse type increase with an excitation period of 60 min was simulated to imitate a *B.cinerea* infection.

A major issue with respect to the implementation of VOC-based crop monitoring is the spatial distribution of VOCs inside a large-scale greenhouse. Our model predictions for large-scale greenhouses relied on the assumption that the air in the greenhouse is perfectly mixed; an assumption often made in classical greenhouse studies (Roy, Boulard, Kittas, &

Table 4 – Predicted baseline and *Botrytis cinerea*-induced increase in concentration of the volatile organic compounds (Z)-3-hexenol, α-pinene, α-terpinene, β-caryophyllene, and methyl salicylate under four different scenarios. The scenarios are presented in Table 3.

	Concentration (mol m^{-3})				
	(Z)-3-Hexenol	α-Pinene	α-Terpinene	β-Caryophyllene	Methyl salicylate
Scenario 1					
Baseline	0	9.995×10^{-8}	1.177×10^{-7}	5.235×10^{-9}	4.719×10^{-7}
Induced	2.030×10^{-7}	1.003×10^{-7}	1.181×10^{-7}	5.254×10^{-9}	4.745×10^{-7}
Increase[a]	∞	0.35	0.34	0.36	0.55
Scenario 2					
Baseline	0	9.995×10^{-8}	1.177×10^{-7}	5.235×10^{-9}	4.719×10^{-7}
Induced	2.030×10^{-6}	1.008×10^{-7}	1.189×10^{-7}	5.301×10^{-9}	4.875×10^{-7}
Increase[a]	∞	0.85	1.02	1.26	3.31
Scenario 3					
Baseline	0	1.662×10^{-9}	1.978×10^{-9}	8.705×10^{-11}	7.912×10^{-9}
Induced	1.226×10^{-8}	1.665×10^{-9}	1.984×10^{-9}	8.736×10^{-11}	7.999×10^{-9}
Increase[a]	∞	0.18	0.30	0.36	1.10
Scenario 4					
Baseline	0	1.662×10^{-9}	1.978×10^{-9}	8.705×10^{-11}	7.912×10^{-9}
Induced	1.226×10^{-7}	1.697×10^{-9}	2.033×10^{-9}	9.021×10^{-11}	8.783×10^{-9}
Increase[a]	∞	2.11	2.78	3.63	11.01

a Increase in %.

Table 5 – Detectability of the *Botrytis cinerea*-induced increase in concentration of (Z)-3-hexenol, α-pinene, α-terpinene, β-caryophyllene, and methyl salicylate under scenarios 1, 2, 3 and 4. An instrument with an accuracy of 10% relative standard deviation and a detection limit of 1 nmol m^{-3} is used as a reference. The increase in concentration of the compounds is presented in Table 4. Scenarios are presented in Table 3.

Compound	Detectability (accuracy/detection limit)			
	Scenario 1	Scenario 2	Scenario 3	Scenario 4
(Z)-3-Hexenol	+/+	+/+	+/+	+/+
α-Pinene	−/+	−/+	−/+	−/+
α-Terpinene	−/+	−/+	−/+	−/+
β-Caryophyllene	−/+	−/+	−/−	−/−
Methyl salicylate	−/+	−/+	−/+	+/+

Wang, 2002). However, VOC concentrations will probably show spatial variation since the air in a large-scale greenhouse is never perfectly mixed. These variations are influenced by the characteristics of a greenhouse, such as the ventilation system and the temperature distribution. Thus concentrations can show a local variation and might be lower or higher than the ones predicted in this study. Furthermore, *B. cinerea* infections are likely to occur in patches. This will also affect the spatial distribution of VOCs and result in an increase in local concentrations. Currently, computational fluid dynamics (CFD) has been acknowledged as an appropriate tool to calculate airflow distributions in greenhouses (e.g. Campen & Bot, 2003). CFD may also be useful to predict airflow distributions and concentrations of VOCs in greenhouse air which then might help to predict the best locations for air sampling.

5. Conclusion

Based on model predictions, an instrument with an accuracy of 10% relative standard deviation and a detection limit of 1 nmol m^{-3}, would be able to detect the *B. cinerea*-induced increase in concentration of methyl salicylate in a large-scale tomato production greenhouse when: (a) windows are fully opened, and (b) the increase in emission continues for at least 1 h, and (c) 5% of the plants are infected. The *B. cinerea*-induced increase in concentration of methyl salicylate is also detectable when (a) windows are closed, and (b) the increase in emission continues for at least 6 h, and (c) 5% of the plants are infected.

The *B. cinerea*-induced increase in concentration of (Z)-3-hexenol is detectable under all scenarios. However, it is expected that, besides infected plants, many additional sources of (Z)-3-hexenol exist including plant debris and nearby field crops especially upon harvest or stress. Plant debris is nearly always present in greenhouses, and harvest and/or stress of nearby crops is extremely difficult to predict. The *B. cinerea*-induced increases in concentration of α-pinene, α-terpinene and β-caryophyllene are probably undetectable in a large-scale tomato production greenhouse. Therefore, the current study suggests that research should focus on the detection of methyl salicylate to indicate *B. cinerea* infections in large-scale tomato production greenhouses.

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